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Large niche differences emerge at the recruitment stage to stabilize grassland coexistence

Chengjin Chu[1] and Peter B. Adler[2]

Department of Wildland Resources and the Ecology Center, Utah State University, Logan, Utah 84322 USA

Abstract. Niche differences and average fitness differences jointly determine coexistence. However, little empirical information about the magnitude of these two mechanisms is available. Using multispecies population models fit to long-term demographic data for common, co-occurring species in five grassland and shrubland plant communities in western North America, we estimated the strength of stabilizing niche differences and average fitness differences. In all five communities, both pairwise and full community comparisons showed evidence for strong stabilizing mechanisms and relatively small average fitness differences. For a total of 17 species pairs, a measure of niche differences based on simulations of invasion growth rates ranged from 0.59 to 0.93 with a mean of 0.81, where 0 indicates complete niche overlap and 1 indicates zero niche overlap. A corresponding measure of average fitness differences ranged from 1.02 to 2.54 with a mean of 1.53, where 1 indicates identical fitness and a value of 2 indicates a fourfold difference in sensitivity to competition. Comparisons of full communities displayed similar patterns: niche differences ranged from 0.58 to 0.69 with a mean of 0.64, and the average fitness differences ranged from 1.42 to 1.63 with a mean of 1.47. In almost every case, the stabilizing mechanisms were much stronger than minimally necessary to prevent competitive exclusion. Considering that all but one of the species we studied are perennial grasses, which are often grouped in the same functional type, the magnitude of these niche differences is surprising. In all five communities, differences between intra- and interspecific effects at the recruitment stage contributed far more to stabilization than interactions affecting growth and survival. Our results indicate that for these abundant, co-occurring species (1) dynamics are far from neutral, with strong niche differences and weak fitness differences combining to stabilize coexistence, and (2) processes operating at early life stages account for a large proportion of the stabilizing effect. Given the limitations of our inductive approach, both these findings represent hypotheses in need of experimental testing.

Key words: biodiversity maintenance; community ecology; demography; fitness differences; grassland; individual-based model; integral projection model; invasion growth rate; niche differences; positive interactions; species coexistence; western United States.

Introduction

Understanding the processes that allow multiple species to coexist is a fundamental problem in community ecology (Gause 1934, Chesson 2000, Bell 2001, Hubbell 2001, Levine and HilleRisLambers 2009, Adler et al. 2010, Mangan et al. 2010, Chisholm and Muller-Landau 2011). Since the formulation of the competitive exclusion principle by Gause (Gause 1934, MacArthur and Levins 1967, Abrams 1983), a variety of coexistence mechanisms have been discovered and investigated, including resource partitioning (Tilman 1982, Silvertown et al. 1999, McKane et al. 2002), frequency-dependent predation (Janzen 1970, Connell 1971, Roughgarden and Feldman 1975, Packer and Clay 2000, Ishii and Shimada 2012), and spatial and temporal environmental heterogeneity (Whittaker et al. 1973, Chesson and Warner 1981, Chesson 2000). Each of these mechanisms focuses on one dimension of niche differentiation. For example, species can coexist stably when they rely on different resources (McKane et al. 2002), are limited by different enemies (Sedio and Ostling 2013), germinate in different years (Angert et al. 2009), or are favored on different microsites (Sears and Chesson 2007). Most coexistence studies have relied on qualitative tests for the presence of one of these coexistence mechanisms, rather than attempts to quantify a mechanism's strength (e.g., Angert et al. 2009). The emphasis on individual mechanisms and qualitative tests has meant a lack of research on the cumulative strength of all possible coexistence mechanisms operating simultaneously.

The neutral theory of biodiversity proposed by Bell (2001) and Hubbell (2001) highlighted this gap in understanding. In the absence of empirical evidence for strong stabilization resulting from all possible coexistence mechanisms, neutral theory proposed that biodiversity is

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[1] Present address: School of Life Sciences, Sun Yat-sen University, Guangzhou, China, 510275.

[2] Corresponding author. E-mail: peter.adler@usu.edu

maintained by ecological equivalence among species. The debate about neutral theory's ability to explain macroscopic patterns of diversity dominated much of the past decade in community ecology (McGill 2003, Volkov et al. 2003, Wootton 2005, Dornelas et al. 2006, Ricklefs and Renner 2012). Evidence that both niche and neutral processes can generate identical community-level patterns (Chave et al. 2002, Purves and Pacala 2005, Chisholm and Pacala 2010) has shifted the focus of the debate, but the original question remains: How important are niche differences, relative to ecological equivalence, in maintaining species diversity?

Recent syntheses of coexistence theory have enabled more sophisticated answers to the niche vs. neutral question. Chesson (2000) established a comprehensive framework for the maintenance of species diversity, in which all possible coexistence mechanisms are grouped into either stabilizing or equalizing forces. Stabilizing niche differences, such as resource partitioning or host-specific natural enemies, cause species to limit themselves more than they limit heterospecifics, buffering each species against extinction by giving it an advantage when it falls to low abundance. Equalizing mechanisms, which can arise from differences in the ability to take up a shared resource or tolerate generalist herbivores, reduce competitive differences among species, slowing the time to competitive exclusion. We refer to these intrinsic competitive differences as average fitness differences. In the absence of stabilizing mechanisms, the species with the highest average fitness would exclude all others. This measure of average fitness, based on an abstraction of dynamics in a world with no niche differences, differs from the traditional measure of individual evolutionary fitness (Chesson 2000, Adler et al. 2007*a*, Carroll et al. 2011, Lankau 2011, Narwani et al. 2013). For stable coexistence, the strength of stabilizing niche differences must overcome the average fitness differences (Chesson 2000, 2013, Adler et al. 2007*a*). Adler et al. (2007*a*) pointed out that a neutral model is a limiting case with zero niche differences and zero average fitness differences. Rather than asking whether communities are structured by niche or neutral processes, a more informative question concerns the role of niche differences and average fitness differences (Adler et al. 2007*a*). Does the high diversity of natural plant communities reflect niche differences overcoming strong average fitness differences, weak niche differences overcoming small average fitness differences, or some other combination?

Understanding the strength of niche differences and average fitness differences is not only critical for assessing theory (Chesson 2000, 2013, Adler et al. 2007*a*, Levine et al. 2008, Narwani et al. 2013, Godoy et al. 2014), but also has important applications in conservation and management. Applied ecology has long recognized that competitive interactions can play an important role in regulating the success and impact of invasive species (e.g., Levine et al. 2004), determining the persistence and diversity of inferior competitors (e.g., Collins et al. 1998), and mediating the effects of global change (Suttle et al. 2007, Gilman et al. 2010). Less appreciated is the influence of niche differences and average fitness differences on the strength of competitive interactions. Stronger niche differences mean weaker plant–plant interactions, while stronger average fitness differences mean greater asymmetry in competitive impacts. For example, an invader which occupies a unique niche with respect to resident species is more likely to successfully invade but less likely to greatly impact resident species (MacDougall et al. 2009). Similarly, in a community characterized by strong niche differences, the indirect effects of climate change mediated by competitive interactions will be weak (Adler et al. 2012).

Quantifying niche differences and average fitness differences in natural or experimental communities presents serious challenges. For some simple models, analytical expressions for niche differences and average fitness differences are known (Chesson 2013, Godoy et al. 2014). More flexible methods for estimating niche differences in experimental or natural communities focus on invasion growth rates, the long-term, low-density growth rate of a focal species surrounded by resident species near their equilibrium abundances. For stable coexistence, each species must have a positive invasion growth rate, which allows it to recover from a perturbation to low density (Chesson and Ellner 1989, Ellner 1989, Chesson 1994). Adler et al. (2007, 2010) and Carroll et al. (2011) have proposed methods to estimate niche differences and average fitness differences based on invasion growth rates measured or simulated under a variety of specific conditions. These approximations make it possible to quantify niche differences and average fitness differences.

Given the very recent development of these approaches, it is no surprise that few empirical studies have estimated niche differences and average fitness differences. Using experimental assemblages of annual plants, Levine and HilleRisLambers (2009; J. M. Levine and J. HilleRisLambers, *unpublished data*) showed that co-occurring species were characterized by strong niche differences and relatively small average fitness differences, while species that failed to persist generally suffered large average fitness disadvantages. Narwani et al. (2013) studied pairs of freshwater green algae in lab microcosms and found a range of mostly weak niche differences that were strong enough to overcome modest average fitness differences in some, but not all, cases. In contrast, Adler et al. (2010), simulating multispecies population models parameterized with long-term observational data from a sagebrush steppe plant community, demonstrated that niche differences were very strong and average fitness differences very weak. In fact, the niche differences were much stronger than necessary to stabilize coexistence. Adler et al. (2010) and Levine and HilleRisLambers (2009) may have found stronger niche differences because their field data captured more potential coexistence mechanisms than the short-term, lab system of Narwani

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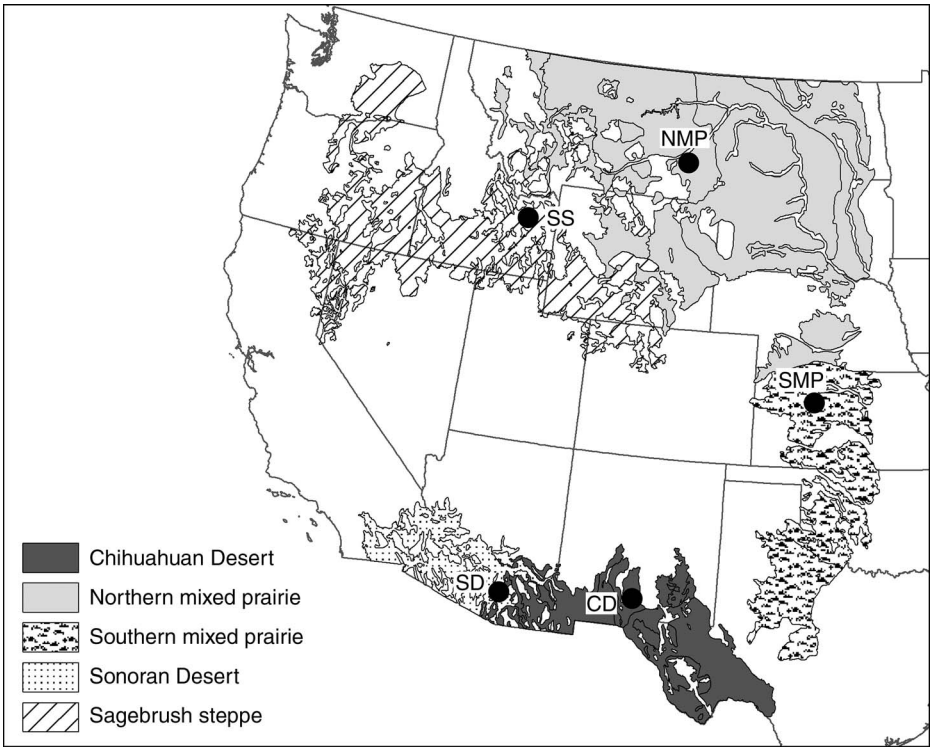


FIG. 1. The location (within the United States) of our five study sites (black circles) and the vegetation types they represent: Sonoran Desert, Arizona (SD), sagebrush steppe, Idaho (SS), southern mixed prairie, Kansas (SMP), northern mixed prairie, Montana (NMP), and Chihuahuan Desert, New Mexico (CD).

et al. (2013). Stronger generalizations will require quantifying niche differences and average fitness differences for many more communities representing a range of taxonomic groups and habitat types.

Although our focus on overall niche differences and average fitness differences lends itself to a phenomenological approach, understanding the role of particular coexistence mechanisms remains an important goal. Fortunately, approaches for quantifying niche differences can provide some inferences about coexistence mechanisms. For example, different stabilizing mechanisms may operate at different life stages (Moll and Brown 2008). By quantifying species interactions separately for survival, growth, and recruitment, and then running simulations focused on the role of each of these life stages, Adler et al. (2010) showed that the recruitment process contributed much more to overall niche differences than did survival or growth. That insight helps narrow the field of potentially important coexistence mechanisms. While resource competition might have stronger effects on survival and growth, we might expect natural enemies, and pathogens in particular, to mediate coexistence via recruitment (e.g., Bagchi et al. 2014). Of course, we cannot generalize about the primary role of recruitment-related processes in stabilizing coexistence from an isolated case study.

The maturation of contemporary coexistence theory (Chesson 2000, 2013, Carroll et al. 2011, HilleRisLambers et al. 2012) has moved these questions into the empirical realm. We now know how to test for niche and fitness differences, we just need to conduct the appropriate experiments and analyses using data from natural communities. Here, we address this need by updating and extending the analysis of Adler et al. (2010) to an additional four data sets from arid to subhumid plant communities in western North America. These data sets, consisting of annual, mapped censuses of individual plants in 1-m^2 quadrats, combine long-term demographic data with neighborhood-scale spatial resolution. As a result, they provide an unprecedented opportunity to explore general patterns of coexistence. Our objectives were to (1) compare the magnitude of niche differences and average fitness differences among the common, co-occurring species in each community, and (2) quantify the contributions of survival, growth, and recruitment processes to niche differences.

MATERIALS AND METHODS

Summary of study sites and species selection

Many range experiment stations in the western United States began mapping permanent quadrats in the early 20th century, and continued annual censuses for decades. Here, we focus on five long-term data sets

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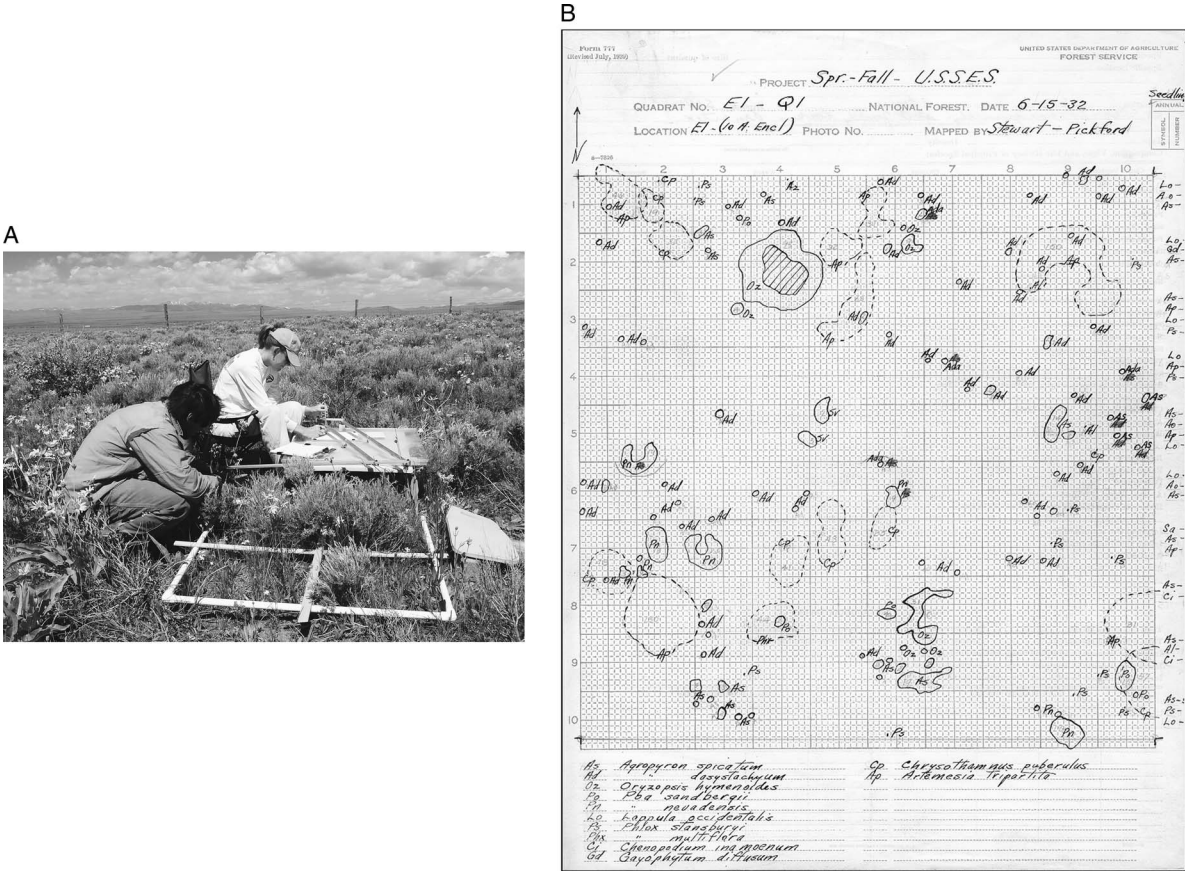


FIG. 2. (A) A contemporary pantograph in use at the sagebrush steppe site (photo credit: Peter Adler) and (B) an example of a historical data sheet from the same site. In (B), mapped polygons with indeterminate shape represent basal cover for grasses and canopy cover for shrubs, and points represent forbs.

(Fig. 1), four of which have been distributed publicly (Adler et al. 2007*b*, Zachmann et al. 2010, Anderson et al. 2011, 2012). At each study site, all individual plants within each 1-m^2 quadrat were identified and mapped annually using a pantograph (Hill 1920; Fig. 2). Mapped polygons represent basal cover for grasses and canopy cover for shrubs (Fig. 2).

To fit our population models, we needed to study species abundant enough to provide a large sample size. Furthermore, to describe species interactions, we needed to select species that co-occurred together across many years. Thus, our analysis focused on the niche differences and average fitness differences of the most common, co-occurring species at each site. Because we studied species that appear to be coexisting, we expected to find that niche differences would be strong enough to offset fitness differences. But we had no expectations about the absolute magnitudes of these differences. Constraining niche differences to be stronger than fitness differences still leaves room for tremendous variation in their potential values.

To select species for our analysis, we first identified species that occurred in at least 20% of years. Then we used nonmetric multidimensional scaling (NMDS) to identify quadrats with a similar composition of these candidate species. Based on the degree of aggregation of quadrats on the NMDS plot, we selected quadrats and species for our full analysis. We also confirmed these selections with local personnel from each study site. We further describe each study site, and our criteria for quadrat and species selections.

Sonoran Desert, Arizona.—The Sonoran Desert (SD) data set comes from 178 permanent quadrats located on semidesert grasslands at the Santa Rita Experimental Range (31°50′ N, 110°53′ W; elevation 1150 m), Arizona (Anderson et al. 2012). During the sampling period, mean annual precipitation was 450 mm, with monthly means ranging from 5 mm in May to 107 mm in July, and mean annual temperature was 16°C, ranging from 8°C (January) to 26°C (July). These quadrats were mapped annually from 1915 to 1933 in most cases. Species composition varies across the broad range of soils and topography that occur within the semidesert grasslands. For our present analysis, we chose 32 quadrats dominated by black grama *Bouteloua eriopoda* (BOER) and *Bouteloua rothrockii* (BORO).

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Sagebrush steppe, Idaho.—The sagebrush steppe (SS) data set comes from the U.S. Sheep Experiment Station, currently a U.S. Department of Agriculture (USDA) Agricultural Research Service field station, located 9.6 km north of Dubois, Idaho (44.2° N, 112.1° W; elevation 1500 m). Twenty-six quadrats were established between 1926 and 1932 (Zachmann et al. 2010). During the period of sampling (1926–1957), mean annual precipitation was 270 mm, with monthly means ranging from 16 mm (March) to 42 mm (June), and mean annual temperature was 6°C, ranging from −8°C (January) to 21°C (July). The vegetation is dominated by the shrub *Artemisia tripartita* (ARTR) and the C_3 perennial bunchgrasses *Pseudoroegneria spicata* (PSSP), *Hesperostipa comata* (HECO), and *Poa secunda* (POSE).

Southern mixed prairie, Kansas.—The southern mixed prairie (SMP) data set represents the pioneering work of Albertson and colleagues (Albertson and Tomanek 1965), who established 51 permanent quadrats inside and outside livestock exclosures near Hays, Kansas (38.8° N, 99.3° W; elevation 650 m). Mean annual precipitation was 580 mm, ranging from 10 mm in January to 103 mm in June, and mean annual temperature was 12°C, ranging from −2.0°C (January) to 27°C (July). These quadrats were mapped from 1932 to 1972 at the end of each growing season (Adler et al. 2007*b*). Variation in soil depth and texture creates distinct plant communities. Many of the quadrats were located on shortgrass communities dominated by *Bouteloua gracilis* and *Buchloë dactyloides*. We had difficulty working with these species, which can be hard to distinguish in the field in the absence of inflorescences, and were often mapped inconsistently. Therefore, we focused on seven quadrats dominated by *Bouteloua curtipendula* (BOCU), *Bouteloua hirsuta* (BOHI), and *Schizachyrium scoparium* (SCSC).

Northern mixed prairie, Montana.—The northern mixed prairie (NMP) data set comes from the Fort Keogh Livestock and Range Research Laboratory, another USDA Agricultural Research Service field station. The study site is located on alluvial plains near the Tongue River (46°19′ N, 105°48′ W; elevation 720 m). Mean annual precipitation was 343 mm and monthly means ranged from 7 mm in February to 74 mm in June. Mean annual temperature was 8°C, ranging from −7°C in January to 25°C in July. Forty-four quadrats, distributed across six pastures assigned to one of three grazing intensities, were mapped annually (with some exceptions) from 1932 through 1945 (Anderson et al. 2011). Our analysis includes 19 quadrats in which *B. gracilis* (BOGR), *H. comata*, *Pascopyrum smithii* (PASM), and *P. secunda* were prevalent.

Chihuahuan Desert, New Mexico.—The last (CD) data set consists of 76 permanent quadrats located in Chihuahuan Desert plant communities at the Jornada Experimental Range (32.62° N, 106.67° W; elevation 1260 m), New Mexico. Mean annual precipitation was 264 mm; monthly means ranged from 6 mm in April to 48 mm in August, and mean annual temperature was 14°C, ranging from 4°C in January to 26°C in July. Quadrats were established from 1915 to 1920 and mapping was conducted annually until the 1960s (it continues on roughly 5-yr intervals to the present). Due to the changes in mapping frequency in the 1960s, and the effects of a severe drought in the 1950s that triggered a conversion of many quadrats from grassland to shrubland, we analyzed data for the period from 1915 to 1950. We selected 48 quadrats in which both *B. eriopoda* and *Sporobolus flexuosus* (SPFL) were common (Wright 1972, Wright and van Dyne 1976).

Analysis overview

Our approach has five steps (Fig. 3): (1) extract demographic data from the annual quadrat maps (arrow 1 in Fig. 3), (2) fit statistical models for vital rates (survival, growth, and recruitment; arrow 2-1 for survival and growth, and 2-2 for recruitment), (3) construct two related multispecies population models for each community, an individual-based model (IBM) (arrow 3-1) and an integral projection model (IPM; arrow 3-2), (4) simulate the models to estimate invasion growth rates in the presence of stabilizing niche differences (arrow 4-1), and (5) simulate the models with perturbed parameters to remove stabilizing mechanisms (arrow 4-2).

Extracting demographic data from mapped quadrats

We used a computer algorithm to track the identity of individual genets based on their spatial locations in permanent quadrats (arrow 1 in Fig. 3; Lauenroth and Adler 2008, Chu et al. 2014). Due to mapping error and the potential for herbaceous perennials to move short distances via resprouting, we begin our tracking algorithm by imposing a 5-cm buffer around each polygon (Fair et al. 1999, Lauenroth and Adler 2008, Chu et al. 2014). We tested the sensitivity of our results to this buffer width, as reported in Appendix A. After adding the buffer area to all polygons of the focal species in year $t - 1$, we then calculated the overlap of each of these polygons with each individual polygon occurring at year t. If the year t individual did not overlap with any conspecific polygon from the previous year, it was labeled as a new recruit. Otherwise, the individual was considered a survivor and inherits the identity of the polygon with which it shared the greatest overlap area. Our approach allows genets to fragment and/or coalesce over the study period.

For the purpose of parameterizing our models, we chose to represent each genet in each year as a circle with an area equal to the sum of all the polygons that compose the genet and a location corresponding to the polygons' centroid. Separate genets can have similar centroids when their constituent polygons are interspersed. Very small plants were originally mapped as points; we represented these as squares with an area of

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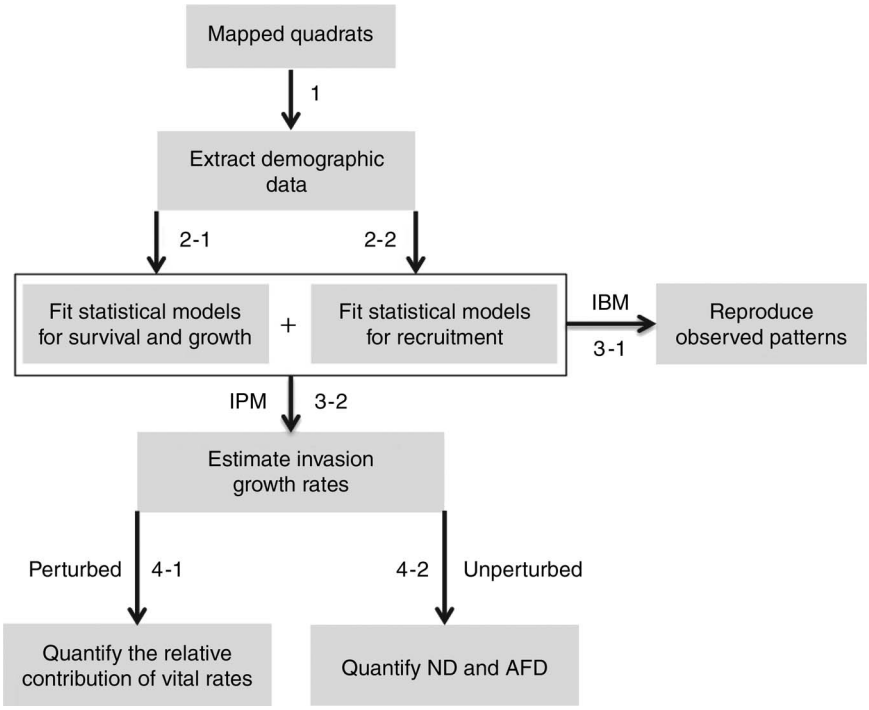


FIG. 3. A workflow of the analysis. Models include individual-based (IBM) and integral projection (IPM), differences include niche (ND) and average fitness (AFD). Arrow numbers correspond to those given in *Materials and methods: Analysis overview*.

0.25 cm^2. The distance between two genets was defined as the distance between their centroids.

Fitting statistical models for survival, growth, and recruitment

Statistical modeling of survival and growth.—We model the survival probability and growth of an individual genet as a function of genet size, the neighborhood-scale crowding experienced by the genet from both conspecific and heterospecific genets, temporal variation among years, and permanent spatial variation among groups of quadrats; here, "group" means a set of nearby quadrats located within one pasture or grazing exclosure (arrow 2-1 in Fig. 3; Adler et al. 2010, 2012).

We assumed that the crowding experienced by a focal genet depends on the distance, d, to each neighbor genet and the neighbor's size, u

$$w_{ijm,t} = \sum_k e^{-\delta_{jm} d^2_{ijkm,t}} u_{km,t} \quad (1)$$

where $w_{ijm,t}$ is the crowding that genet i of species j in year t experiences from neighbors of species m, δ_{jm} determines the spatial scale over which neighbors of species m exert influence on a genet of species j ($m = j$ represents intraspecific crowding), k indexes all the focal genet's neighbors of species m at time t, and $d_{ijkm,t}$ is the distance between genet i in species j and genet k in species m. The use of squared distances implies a Gaussian interaction kernel.

To fit the δ values, which describe the distance-decay of interactions, we began with a series of simple, single-species models of survival and growth. These models included genet size and different combinations of spatially and temporally varying parameters, all treated as fixed effects. We fit each of these models with a wide range of δ values, and used the Akaike information criterion (AIC) to identify the optimal value of δ for survival and growth, separately.

Once we had determined the intraspecific δ's, we evaluated three possibilities for assigning the interspecific δ's. First, we considered an effects model, where each species has a characteristic zone of influence: an individual of a given size will affect all its neighbors similarly, regardless of the neighbor's species. In other words, the value of δ in Eq. 1 depends on the species identity of the neighbor, not the focal individual (i.e., $\delta_{jm} = \delta_m$, where δ_m is the value from the selected single-species model for species m). Second, we considered a response model, where the value of δ in Eq. 1 depends only on the identity of the focal individual (i.e., $\delta_{jm} = \delta_j$). Third, we examined an intermediate model, i.e., the interaction model, which assumes that both effects and responses are relevant. In this case, the value of δ is an average of the single-species values for the focal species and the neighbor. To compare these three scenarios, we fit survival and growth regressions for each species using multispecies neighborhoods, with either the effects,

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response, or interaction models defining the interspecific δ's based on the single-species values. We found that the effects model was superior more often than the other two models, although for most species the differences were small (Appendix B). Therefore, we used the effects model for all species for interspecific interactions.

We modeled the survival probability, S, of genet i in species j and group g from time t to $t + 1$ as

$$\text{Logit}(S_{ijg,t}) = \gamma^S_{j,t} + \varphi^S_{jg} + \beta^S_{j,t}u_{ij,t} + \boldsymbol{\omega}^S_j\mathbf{w}_{ij,t} \qquad (2)$$

where γ is a time-dependent intercept, φ is the coefficient for the effect of group, β is the coefficient that represents the effect of genet size, u, on survival, and $\boldsymbol{\omega}$ is a vector of per capita interaction coefficients which determine the impact of crowding, $\mathbf{w}$, by each species on the focal species. Fitting this model to the data included estimation of the interaction coefficients.

Our model for changes in genet size from time t to $t + 1$, conditional on survival, has a similar structure

$$u_{ijg,t+1} = \gamma^G_{j,t} + \varphi^G_{jg} + \beta^G_{j,t}u_{ij,t} + \boldsymbol{\omega}^G_j\mathbf{w}_{ij,t} + \varepsilon^G_{ij,t} \qquad (3)$$

where G refers to growth. Additionally, to capture nonconstant error variance in growth, we followed previous studies (Ellner and Rees 2006, Adler et al. 2010) in modeling the variance ε about the growth curve (Eq. 3) as a nonlinear function of predicted genet size

$$\text{Var}(\varepsilon^G_{ij,t}) = ae^{bu_{ij,t+1}} \qquad (4)$$

where a and b are fitted coefficients. Note that a different set of per capita interaction coefficients, $\boldsymbol{\omega}$, were fit for each vital rate, and they can take positive or negative values. In fact, we found that positive interactions were common, though typically they were weak. Because the approaches we used for calculating niche differences and average fitness differences cannot accommodate positive interactions, we fit a version of the regressions with strong priors, which restricted the per capita interaction coefficients to be negative, implying competition (see the section *Excluding positive interactions from vital rates*). Comparison of these strong prior models with diffuse prior models (positive interactions permitted) indicated little loss of model fit and only small changes in equilibrium abundances (Appendix C). Therefore, we used models with only negative species interactions for our simulations. We consider the implications of this choice in *Discussion*.

While Eqs. 2 and 3 allow the intercept, γ, size effect, β, and species interaction coefficients, $\boldsymbol{\omega}$, to vary through time, we let the observed data determine whether we modeled these coefficients as constant or time varying. We fit a number of alternative multispecies models in a generalized linear models (GLM) framework, treating all time-varying parameters as fixed effects (fitting a separate parameter for each effect for each year) and using AIC to identify the model with the highest predictive accuracy for each species. We then fit a mixed-effects version of this model, treating all year and spatial location effects as random effects. For example, when the interaction between genet size and year was significant in the fixed-effects model, we added the random effect of year on genet size in the corresponding mixed-effects model. We fit these models using lmer (for growth) and glmer (for survival) in R package lme4 (version 1.1-6; R Development Core Team 2014). When we refit these models using Bayesian methods to exclude positive interactions, we retained the model structure selected at this stage (see *Excluding positive interactions from vital rates*).

Statistical modeling of recruitment.—The mapped data do not allow us to determine which recruits were produced by which potential parent genets. Therefore, we model recruitment at the quadrat level rather than at the individual genet level (arrow 2-2 in Fig. 3). We assume that the number of individuals, y, of species j recruiting at time $t + 1$ in the location q follows a negative binomial distribution

$$y_{jq,t+1} \sim \text{NegBin}(\lambda_{jq,t+1}, \theta) \qquad (5)$$

where λ is the mean intensity and θ is the size parameter. In turn, λ depends on the composition of the quadrat in the previous year

$$\lambda_{jq,t+1} = A'_{jq,t}e^{(\gamma^R_t + \varphi^R_g + \boldsymbol{\omega}^R\sqrt{A'_{q,t}})} \qquad (6)$$

where R refers to recruitment, γ is a time-dependent intercept, φ is a coefficient for the effect of group location, $\boldsymbol{\omega}$ is a vector of coefficients that determine the strength of intra- and interspecific density dependence, and $A'_{jq,t}$ is the effective cover (cm^2) of species j in quadrat q at time t. We modeled the effect of latent effective cover rather than observed cover to account for the possibility that plants outside the mapped quadrat could contribute recruits to the focal quadrat, or interact with plants in the focal quadrat. We estimated effective cover as a mixture of the observed cover, A, in the focal quadrat, q, and the mean cover, $\bar{A}$, across the group, g, in which the quadrat is located

$$A'_{jq,t} = p_jA_{jq,t} + (1 - p_j)\bar{A}_{jg,t} \qquad (7)$$

where p is a mixing fraction between 0 and 1.

Eq. 6 is essentially the Ricker model for discrete-time population growth (Ricker 1954), except that we use the square root of cover in the density-dependent term. We chose not to directly use the Ricker equation as Adler et al. (2010) did, as recruitment models with the square-root transformation had smaller deviance information criterion (DIC) values than the ones without the transformation (Spiegelhalter et al. 2002). Following previous work (Adler et al. 2006, 2010), we treated year and group variables as random factors, allowing intercepts to vary among years and spatial locations.

In our recruitment models, error in measurements of population cover could bias our estimates of intraspecific density dependence (Freckleton et al. 2006). To test the potential impact of such measurement error on our

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parameter estimates, we conducted additional field measurements in the summer of 2013 for four species (*A. tripartita*, *H. comata*, *P. secunda*, and *P. spicata*) from the sagebrush steppe plant community: two different observers mapped the same four quadrats two times each (Appendix D). Using these observations, we estimated the measurement error for each species. We then fit a version of the model that accounted for measurement error. We found that the small degree of measurement error involved in pantograph mapping had only a minor influence on parameter estimates and corresponding community dynamics (Appendix D). Based on this assessment, we decided not to include measurement error in the recruitment models we used for all five plant communities.

Excluding positive interactions from vital rates.—We excluded positive interactions from our final vital rate models (survival, growth, and recruitment) by applying informative priors in a Bayesian computational framework. Specifically, we constrained the interaction coefficients (Eqs. 2, 3, and 6) to values less than 0. We ran Markov Chain Monte Carlo (MCMC) simulations implemented in WinBUGS 1.4 (Lunn et al. 2000) to estimate the model parameters. Each model was run for 20 000 iterations and three chains with different initial values for parameters. We discarded the initial 10 000 samples. Convergence was observed graphically for all parameters and diagnosed using the Brooks-Gelman statistic (Brooks and Gelman 1998). The remaining converged MCMC samples were thinned every 50 to reduce autocorrelation. The fitted models showed that large genets usually had high survival rates but low relative growth, while small genets had low survival rates but high relative growth (Appendices E and F). Recruitment was low in most years and quadrats, with occasional years of high recruitment (Appendix G). Detailed statistical results for all three vital rates are presented in Appendix H.

Building multispecies IBM/IPMs

The vital rate regressions described are the building blocks for two similar multispecies population models. We used spatially explicit, individual-based models (IBMs) to confirm that our models could reproduce the observed dynamics during the period of historical sampling at each site, and we used spatially implicit, integral projection models (IPMs) to explore the asymptotic behavior of the modeled populations (i.e., to estimate niche differences and average fitness differences).

Individual-based model.—Based on the statistical models for each of the vital rates, we constructed individual-based models (IBM) as follows (arrow 3-1 in Fig. 3). To simulate time series of abundances in each quadrat, we initialized the IBM with the actual conditions (plant sizes and locations) observed in the first year the quadrat was recorded. We then projected the model forward in time, drawing from the year-specific parameters in chronological order, and applying location-specific parameters as appropriate for each quadrat. These year- and location-specific parameter values correspond to the year and group random effects included in the statistical models for vital rates. For these simulations, we used absorbing boundaries. Where data gaps in the observed time series occurred, we reinitialized the IBM with the plant sizes and locations observed in the year following the gap. For each quadrat, we ran 50 replicate simulation runs and averaged cover and density across runs for each species. To simulate data for analysis of spatial distributions, we ran 50 simulations of 100 time steps for an average location, and saved plant sizes and spatial coordinates every 10 time steps after a 50-time-step burn-in period. We then calculated Ripley's K and cross-K functions to describe intra- and interspecific spatial patterns (Ripley 1977), respectively, using the spatstat package in R (Baddeley and Turner 2005).

Integral projection model.—The demographic stochasticity present in the IBM complicates analyses of invasion growth rates and equilibrium abundances. Therefore, we built an environmentally stochastic but demographically deterministic version of the model using an IPM (arrow 3-2 in Fig. 3; Adler et al. 2010). In our IPM, the population of species j is represented by a density function $n(u_j,t)$ that gives the density of genets of size u at time t, with genet size on a natural-log scale, i.e., $n(u_j,t)du$ is the number of genets whose area (on an arithmetic scale) is between $\exp(u_j)$ and $\exp(u_j + du)$. The density function for size v at time $t + 1$ is given by

$$n(v_j, t+1) = \int_{L_j}^{U_j} k_j(v_j, u_j, \bar{\mathbf{w}}_j(u_j))n(u_j, t) \tag{8}$$

where the kernel $k_j(v_j,u_j,\bar{\mathbf{w}}_j)$ describes all possible transitions from size u to v and $\bar{\mathbf{w}}_j$ is a vector whose elements are the average crowding experienced from each of the other species by an individual of size u_j of species j (we will describe how $\bar{\mathbf{w}}_j$ is calculated). The integral is evaluated over all possible sizes from a lower size limit L to an upper size limit U that extends past the range of observed sizes.

The kernel is constructed from the fitted survival (S), growth (G), and recruitment (R) models

$$k_j(v_j, u_j, \bar{\mathbf{w}}_j) = S_j(u_j, \bar{\mathbf{w}}_j(u_j))G_j(v_j, u_j, \bar{\mathbf{w}}_j(u_j)) + R_j(v_j, u_j, \bar{\mathbf{w}}_j) \tag{9}$$

where S is given by Eq. 2 and G by Eq. 3. The recruitment function, R, defined in Eqs. 5 and 6, gives the number of new recruits produced per quadrat. To incorporate this recruitment function into the IPM, we assumed that fecundity increases linearly with size $R_j(v_j,u_j,\bar{\mathbf{w}}_j) = e^{u_j} R_j(v_j,\bar{\mathbf{w}}_j)$ (Adler et al. 2010, Dalgleish et al. 2011).

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In the vital rate regressions and in the IBM, we calculated a neighborhood crowding index, w_{ij}, unique to each individual i based on the spatial locations and sizes of neighboring plants (Eq. 1). This measure cannot be extended to the IPM, because the IPM is spatially implicit. Therefore, we used an approximation to $\bar{\mathbf{w}}_j$ that captures the essential features of spatially explicit neighborhood interactions (Adler et al. 2010). We found that in both the observed data and IBM simulations, heterospecific individuals were approximately randomly distributed, but conspecific individuals displayed nonrandom, size-dependent spatial patterns: while small genets were randomly distributed, large genets were segregated from each other (Appendix I). The overdispersion of large conspecific genets was introduced into the IPM with a no-overlap rule (Adler et al. 2010, 2012).

IPM simulations

We simulated the IPMs for each community to estimate (1) the equilibrium abundance of each species and (2) each species' invasion growth rate. To simulate equilibrium abundances, we initialized all species in each community with arbitrarily low cover and ran the model for 2000 time steps to allow the community to reach an equilibrium. We calculated the equilibrium cover for each species by averaging over an additional 500 records. To model temporal environmental variation, at each time step we randomly selected a set of estimated survival, growth, and recruitment parameters specific to one of the observed years, i.e., the estimated year-specific random effects.

To quantify invasion growth rates, we simulated the dynamics of a focal species at extremely low abundance in the presence of resident species at their stochastic equilibria. In contrast to the simulation of equilibrium abundances that included all species in each community, we initialized this simulation with all species but one, the "focal" species. We assigned the "resident" species, those present at initialization, with arbitrarily low cover values and ran the models for 1000–4000 time steps to allow the residents to reach an equilibrium in the absence of the focal species. Then we introduced the focal species at a cover of 10^{-6} cm^2. We calculated the invasion growth rate as $\log(A_{t+1}/A_t)$ where A is the areal cover of the focal species. After this calculation, we returned the cover of the focal species to 10^{-6} cm^2 and repeated this short-term invasion experiment 1000 times. The long-term invasion growth rate of the focal species was estimated as the species' mean invasion growth rate over this 1000-yr time period (because we average the log growth rates, this is a geometric mean).

As we describe in the next section, to quantify niche differences and average fitness differences, we repeated these simulations after perturbing model parameters in various ways.

Quantifying niche differences and average fitness differences

Adler et al. (2010) used each species' simulated invasion growth rate as a proxy for stabilizing niche differences (higher invasion growth rates indicate stronger niche differences), and quantified average fitness differences by simulating community dynamics after setting the interspecific interaction coefficients equal to the intraspecific interaction coefficients, essentially making each species' effects on all other species equal to its effect on itself. This approach successfully removed niche differences, but we now understand that it also altered species' sensitivity to competition, an aspect of average fitness differences, confounding the comparison of niche differences and average fitness differences. Since the publication of Adler et al. (2010), two more-rigorous approaches have become available. Chesson (2013) derived analytical expressions for niche differences and average fitness differences in a Lotka-Volterra competition model. We adapted this approach to explore the contributions of each vital rate to coexistence. Carroll et al. (2011) developed an approach to calculate the estimates of niche differences and average fitness differences based on invasion growth rates. We applied this approach as well.

For a two-species Lotka-Volterra competition model, Chesson defined niche overlap as

$$\rho = \sqrt{\frac{\alpha_{ij} \times \alpha_{ji}}{\alpha_{jj} \times \alpha_{ii}}} \tag{10}$$

and average fitness difference as

$$\frac{k_j}{k_i} = \sqrt{\frac{\alpha_{ii} \times \alpha_{ij}}{\alpha_{jj} \times \alpha_{ji}}} \tag{11}$$

where α_{ii} is for intraspecific competition coefficients, and α_{ij} is for interspecific competition coefficients. Although our model differs from the standard Lotka-Volterra model, its phenomenological treatment of competition is similar. Therefore, we used Chesson's expressions (Eqs. 10 and 11) as a basis for exploring the consequences of removing stabilizing niche differences from each community (arrow 4-1 in Fig. 3). Our first step in removing stabilizing niche differences was to disable any fluctuation-dependent coexistence mechanisms implicit in our model (Chesson 2000) by setting all random year effects to zero, resulting in a constant environment model. The stabilizing mechanisms that remain in the constant model are generated by differences in the intra- and interspecific interaction coefficients contained in each vital rate model. We perturbed the interspecific interaction coefficients ($\boldsymbol{\omega}$) to remove the niche difference between each pair of species, while keeping the ratio k_j/k_i unchanged to preserve the average fitness difference. Specifically, for each pair of species i and j in the community, we rescaled the interspecific competition coefficients with the factor

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$$\sqrt{\frac{\alpha_{jj} \times \alpha_{ii}}{\alpha_{ij} \times \alpha_{ji}}}$$

based on Chesson (2013). Because we have a separate matrix of interaction coefficients for each vital rate, applying this perturbation results in three models: no-niche-S, no-niche-G, and no-niche-R correspond to removal of stabilization mechanisms from survival, growth, and recruitment, respectively. Rescaling the interspecific interactions for all three vital rates at once produces a no-niche-RGS model. In this model, all stabilizing mechanisms are absent and the species with the highest average fitness should exclude all others.

We simulated the IPM to estimate equilibrium abundances for each species in the all-niches, constant, no-niche-R, no-niche-S, no-niche-G, and no-niche-RGS models (arrow 4-1 in Fig. 3). We also repeated our simulations of invasion growth rates for all six of these scenarios. By removing niche differences from survival, growth, and recruitment independently, we do not mean to imply that their effects on population size or invasion growth rate are additive. Survival and growth influence the IPM kernel multiplicatively (Eq. 9) and the density-dependent feedbacks make the model nonlinear. These perturbations represent a relative comparison of the impact of niche differences operating within each vital rate process rather than an additive partitioning of their effects.

Carroll et al. (2011) developed an alternative measure of niche differences and average fitness differences that can be applied to either species pairs or to the whole community. While we applied Chesson's approach to evaluate the stabilizing effect of niche differences at the level of vital rates, we applied Carroll's approach to understand niche differences and average fitness differences at the population level (arrow 4-2 in Fig. 3). The approach is based on species' sensitivity (S_i) to competition. S_i is defined as the amount by which a species' intrinsic per capita growth rate is reduced when invading a monoculture of a competing species

$$S_i = 1 - \frac{u_{i,\text{inv}}}{u_{i,\text{int}}}$$

where $u_{i,\text{int}}$ is the per capita growth rate in the absence of competitors and $u_{i,\text{inv}}$ is the per capita growth rate in the presence of resident competitors. For a community with n species, the niche differences (ND) and the average fitness differences (AFD) are quantified as the geometric mean and the geometric standard deviation of S_i, respectively

$$\text{ND} = 1 - \prod_{i=1}^{n} S_i^{1/n} \tag{12}$$

$$\text{AFD} = \exp\left[\left(\overline{\ln(S)^2} - \overline{\ln(S)}^2\right)^{1/2}\right]. \tag{13}$$

An AFD with a value of 1 indicates no fitness difference among species and a value of 2 indicates a fourfold difference in sensitivity to competition. For a given community, species can stably coexist if

$$\text{AFD} \leq \frac{1}{1 - \text{ND}}.$$

This expression was tested numerically for a Lotka-Volterra model and MacArthur's consumer–resource model (Carroll et al. 2011, Narwani et al. 2013), but it is unclear if it works for communities of more than two species and we do not know how precisely it applies to our size-structured population model. Therefore, we treat it as an approximate coexistence criterion to aid in interpretation. Recently, Narwani et al. (2013) applied this method to quantify the niche and average fitness differences of experimental freshwater green algae communities. For communities with more than two species, we decided to calculate ND and AFD for both the full community and for the C_2^n pairwise species subsets of the community (C means combination). We expected the pairwise subsets to yield larger niche differences than the full community analysis because it is easier for species to minimize niche overlap with one competitor than with many competitors.

For the pairwise comparisons, we first simulated the intrinsic per capita growth rate ($u_{i,\text{int}}$) for each species in the absence of the other species, by setting the initial cover for the focal species to 10^{-6}, and the other species to 0. After each time step, we returned the cover of the focal species to 10^{-6}. To account for environmental stochasticity (i.e., the random year effects), we repeated this calculation over 1000 time steps. The per capita growth rate for each time step was calculated as $\log(A_{t+1}/A_t)$, where $A_t = 10^{-6}$. The mean (log) intrinsic per capita growth rate for the focal species was estimated by taking the average over 1000 time steps. Next, we simulated the invasion growth rate ($u_{i,\text{inv}}$) of each species in the presence of each other species in the community. Thus, we had six pairwise comparisons for the SS and NMP study sites, which included four species, three pairwise comparisons for the SMP study site with three species, and one pairwise comparison for the SD and CD study sites, which included only two species. In total, we conducted 17 pairwise comparisons. After we estimated the intrinsic per capita growth rates ($u_{i,\text{int}}$) and invasion growth rates ($u_{i,\text{inv}}$), we calculated the species' sensitivity to competition (S_i), and niche differences and average fitness differences using Eqs. 12 and 13. We followed the same procedure to obtain the community-level niche differences and average fitness differences for each community. The only difference in the community-level case is that $u_{i,\text{inv}}$ comes from simulations of each species in the presence of all other species in the community.

Results

Reproducing historical dynamics

Before relying on our simulation models to explore long-term dynamics, we first show that the fitted

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individual-based models (IBMs) can successfully recover the observed, historical population fluctuations. For all five communities, mean cover and density from quadrat-level simulations closely approximated the observed cover and density means (Fig. 4). IBM simulations also captured essential features of the observed spatial patterns for most species or pairs of species. Spatial patterns for large, conspecific genets tended to be overdispersed in the observed data and in simulations, while comparisons among small genets and all heterospecific genets revealed random distributions (Appendix I).

Stabilizing mechanisms

The statistical models for vital rates demonstrated that intraspecific interaction coefficients were more negative than the interspecific interaction coefficients (Table 1; interaction coefficients for each species are reported in Appendix H). Consistent with the difference between intra- and interspecific interaction coefficients, simulations of the IPMs showed that all species in each community had positive invasion growth rates and reached nonzero equilibrium cover, indicating stable coexistence (Fig. 5).

To provide insight about the nature of the stabilizing mechanisms, we conducted a series of simulations in which we progressively perturbed model parameters. Removal of temporal fluctuations did not greatly alter the invasion growth rates or species' cover at equilibrium (Fig. 5), indicating that the temporal-dependent mechanisms capable of operating in our model contributed little to stabilization. We next removed fluctuation-independent mechanisms by perturbing species interaction coefficients in the growth (no-niche-G), survival (no-niche-S), and recruitment (no-niche-R) models. Removing stabilization mechanisms from recruitment always led to the greatest reduction in invasion growth rates (Fig. 5). In all five communities, the removal of stabilization mechanisms from recruitment actually led to competitive exclusion: *B. eriopoda* in SD, *A. tripartita* and *P. spicata* in SS, *B. hirsuta* in SMP, *B. gracilis*, *H. comate*, and *P. secunda* in NMP, and *S. flexuosus* in CD. Removal of fluctuation-dependent and fluctuation-independent stabilization mechanisms from all three vital rates (i.e., no-niche-RGS models) always led to competitive exclusion: in each community only one species had a positive invasion growth rate (Fig. 5). The dominant species were *B. rothrockii* in SD, *P. secunda* in SS, *B. curtipendula* in SMP, *P. smithii* in NMP, and *B. eriopoda* in CD.

Average fitness differences

Differences in invasion growth rates simulated by the no-niche-RGS provide information about average fitness differences between the superior species and the inferior (excluded) species (Fig. 5). In SD, the difference in invasion growth rates between species *B. rothrockii* and *B. eriopoda* was about 0.40, meaning a population decline rate for the inferior species (*B. eriopoda*) of 40% per year. In SS, the differences between the superior species (*P. secunda*) and the three inferior species (*A. tripartita*, *H. comate*, and *P. spicata*) were 0.051, 0.024, and 0.030, respectively. In SMP, the differences between *B. curtipendula* and the two inferior species (*B. hirsuta* and *S. scoparium*) were 0.37 and 0.42, respectively. In NMP, the differences between *P. smithii* and the three inferior species (*B. gracilis*, *H. comate*, and *P. secunda*) were 0.16, 0.16, and 0.41, respectively. In CD, the difference between the superior species (*B. eriopoda*) and the inferior species (*S. flexuosus*) was 0.20.

A pairwise-species measure of average fitness differences based on Carroll et al. (2011), calculated from simulations of the unperturbed model, provides a unitless comparison between any two species, not just comparisons with the competitive dominant (Fig. 6). In SD and CD, communities in which we modeled only two species, the average fitness differences were 1.64 and 1.30, respectively, indicating a difference between species in sensitivity to competition less than fourfold. For the other three communities, the pairwise comparisons ranged from 1.02 to 2.54 with a mean of 1.59 in SS, from 1.45 to 2.48 with a mean of 1.83 in SMP, and from 1.02 to 1.73 with a mean of 1.34 in NMP. Overall, three species pairs had an average fitness difference larger than 2.0. For the three communities we modeled with more than two species, the community-level average fitness difference was 1.42 for SS, 1.62 for SMP, and 1.42 for NMP.

The balance between niche differences and average fitness differences

We also used Carroll et al.'s (2011) approach to quantify niche differences on a scale from 0 (complete niche overlap) to 1 (no overlap). For the pairwise comparisons, across all communities the niche differences ranged from 0.59 to 0.93, with a mean of 0.81, and most values were larger than 0.7, indicating very little niche overlap. On the community level, the niche difference was 0.58 for SD, 0.59 for SS, 0.66 for SMP, 0.69 for NMP, and 0.68 for CD. The mean of niche differences across communities was 0.64, which was smaller than the mean of pairwise comparisons (0.81). Based on Carroll et al.'s (2011) coexistence criteria, both pairwise comparisons and the community-level comparisons indicate a stable coexistence (Fig. 6), consistent with the simulations of our all-niches models.

Discussion

In a previous analysis of the sagebrush steppe community (Adler et al. 2010), we found strong niche differences acting on weak average fitness differences and evidence that recruitment processes played a primary role in stabilizing coexistence. By applying improved methods to an additional four communities,

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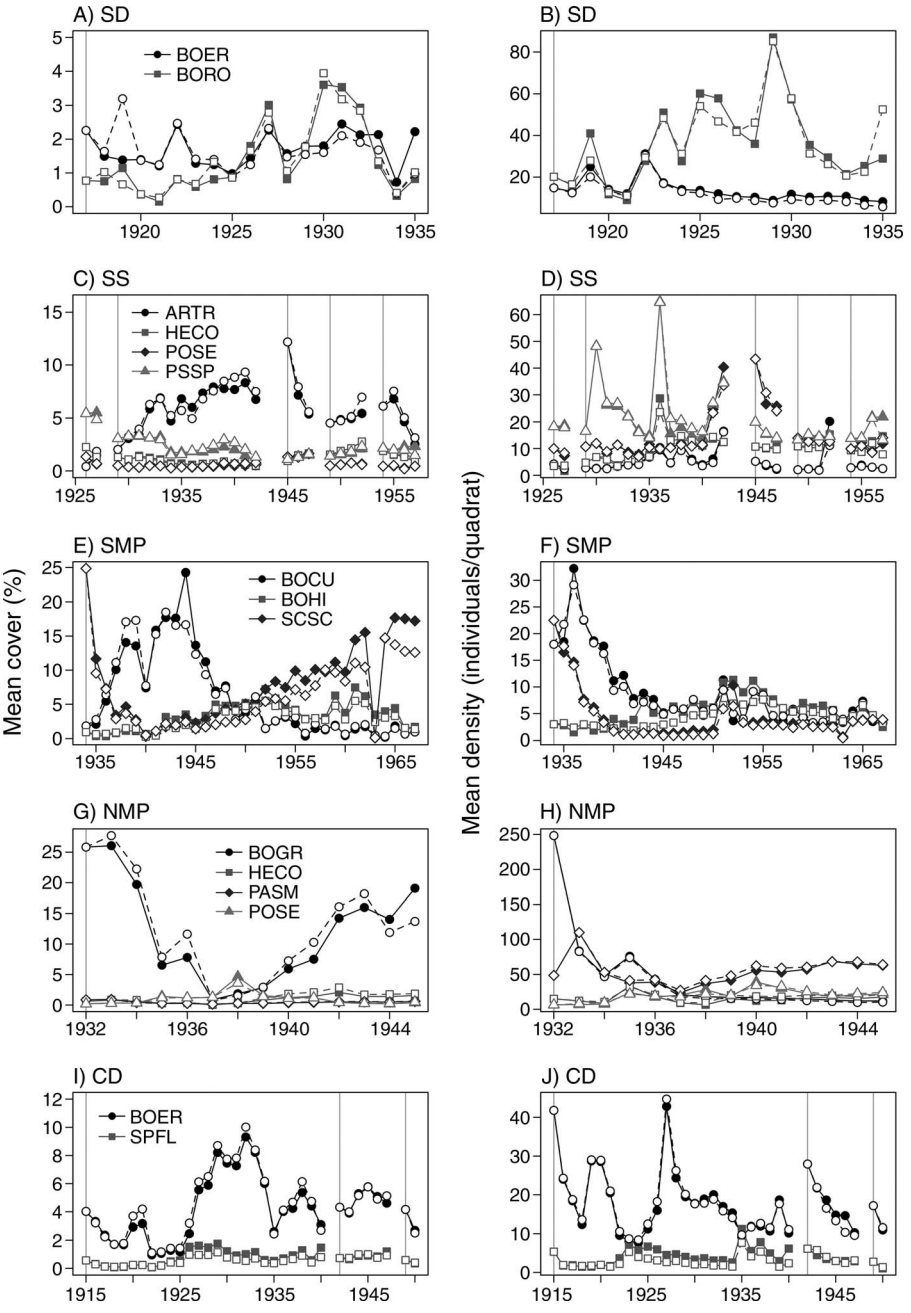


FIG. 4. Observed and predicted abundance. Solid symbols show observed means for cover (A, C, E, G, and I) and density (B, D, F, H, and J) and open symbols show simulated means from IBMs. For each quadrat, the model was initialized with the observed genet size and spatial coordinates at dates indicated by the vertical gray lines and then projected forward until the end of the time series or a missing year. After missing census years, we reinitialized the simulated quadrat with the plant sizes and locations observed in the year following the data gap. Observed means are averaged across quadrats within each year; predicted means are averaged across 50 simulations runs for each quadrat. See Fig. 1 for the abbreviations of the plant communities, and *Materials and methods: Summary of study sites and species selection* for the abbreviations of species names.

along with the original sagebrush data set, we have shown that these results do not represent an unusual case study, but rather appear general. In all five of the arid-to-subhumid grassland and shrubland communities that we studied, coexistence is far from neutral, reflects very strong niche differences, and is most strongly stabilized by recruitment processes. Although the strength of the niche differences we report may appear surprising, our results are consistent with recent research in a variety of systems.

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TABLE 1. The averages of interaction coefficients (ω) for the vital rates for each community.

Community and interaction	Recruitment	Growth	Survival
SD			
Intra	−1.44 (0.8945)	−0.039 (0.03625)	−0.45 (0.1649)
Inter	−0.45 (0.1152)	−0.013 (0.0794)	−0.051 (0.0307)
SS			
Intra	−1.58 (0.4079)	−0.038 (0.01817)	−0.52 (0.2498)
Inter	−0.24 (0.04978)	−0.003 (0.0006767)	−0.01 (0.003678)
SMP			
Intra	−1.07 (0.1194)	−0.0040 (0.001384)	−0.061 (0.02491)
Inter	−0.12 (0.02413)	−0.0018 (0.0006325)	−0.0068 (0.002411)
NMP			
Intra	−2.08 (0.7448)	−0.27 (0.2042)	−0.81 (0.2833)
Inter	−0.15 (0.02659)	−0.010 (0.004134)	−0.014 (0.003433)
CD			
Intra	−1.65 (0.8849)	−0.0093 (0.005251)	−0.22 (0.0628)
Inter	−0.36 (0.2632)	−0.0038 (0.0003395)	−0.014 (0.00097)

Notes: The value in parentheses is the standard error (SE) across species in each community. Interaction coefficients for vital rates for each species are reported in Appendix H. The communities listed (all within the United States) are Sonoran Desert, Arizona (SD), sagebrush steppe, Idaho (SS), southern mixed prairie, Kansas (SMP), northern mixed prairie, Montana (NMP), and Chihuahuan Desert, New Mexico (CD).

Dynamics reflect niches, not neutrality

Communities may deviate from perfect neutrality (or ecological equivalence) in two ways (Adler et al. 2007*a*): individuals may be more limited by interactions with conspecifics than with heterospecifics due to niche differences, which act to prevent competitive exclusion, and species may vary in average fitness, which accelerates competitive exclusion. We quantified niche differences and average fitness differences using two approaches. Our first approach focused on invasion growth rates: the population growth rate of a focal species introduced at low abundance into a community of residents near their equilibrium abundance. In a perfectly neutral model, each species should have an invasion growth rate equal to zero (on a log scale). Every species we modeled had a positive invasion growth, indicating stable coexistence and a departure from perfect neutrality. Finding positive invasion growth rates and evidence for stable coexistence was no surprise, given that we only studied common species that co-occurred for decades. But it is surprising to find evidence for such strong niche differences among perennial grasses, which represent 14 of the 15 species we studied. Ecologists often assign all perennial grasses to one plant functional type, implicitly assuming a common ecological strategy and minimal niche differences. Furthermore, the magnitude of the positive invasion growth rates was impressive, ranging from 0.013 to 1.04 with a mean of 0.42. These growth rates correspond to population doubling times ranging from one to 55 years, with a mean of less than two years. Most of the species we studied are strongly buffered against local extinction.

We also used invasion growth rates to draw inferences about average fitness differences. In the absence of all stabilizing niche differences, differences in invasion growth rates serve as proxies for average fitness differences (Adler et al. 2010). We removed stabilizing niche differences by simulating our models in a constant environment and perturbing the interaction matrices to remove differences between intra- and interspecific effects. As expected, in these no-niche simulations, only one species in each community had a positive invasion growth rate, while the remaining species had negative invasion growth rates leading to the inevitable competitive exclusion. But the magnitude of these growth rates was again more interesting than their sign, ranging from −0.30 to −0.014, with a mean of −0.087. These rates correspond to population half-lives of two to 51 years, with a mean of eight years. Thus, in the absence of stabilizing mechanisms, competitive exclusion would be relatively slow for many of the species we studied.

The second approach we used also revealed strong niche differences and weak average fitness differences (Fig. 6). We followed Carroll et al. (2011) through numerical simulations. Pairwise niche differences ranged from 0.59 to 0.93, while average fitness differences ranged from 1.02 to 2.54. The community-level niche differences ranged from 0.58 to 0.69 and the average fitness differences ranged from 1.42 to 1.63. The modestly lower values for community-level niche differences are expected; it is easier for a species to avoid overlapping in habitat or resource requirements with one competitor than many competitors (Chesson 2000). Stable coexistence requires that

$$\mathrm{AFD} \leq \frac{1}{1-\mathrm{ND}}.$$

Fig. 6 shows that this inequality is not only satisfied for each pair of interacting species and for each community

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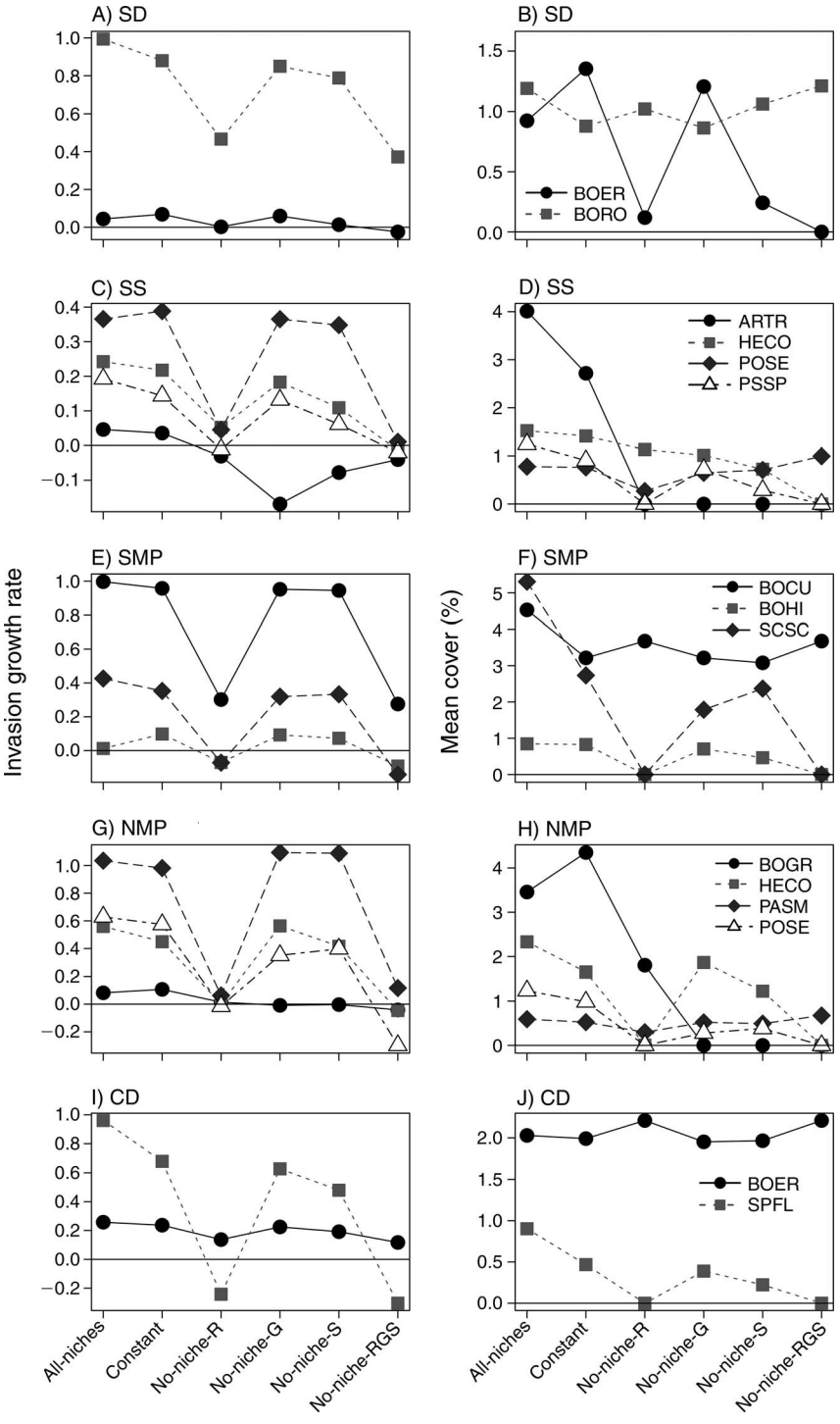


FIG. 5. Invasion growth rates (left column) and mean cover (right column) simulated by the IPM models for each study site. We simulated the IPM model with the estimated parameters (all-niches), after removing random year effects to exclude fluctuation-dependent coexistence mechanisms (constant), and in a constant environment with stabilizing mechanisms removed from recruitment (no-niche-R), growth (no-niche-G), survival (no-niche-S), or all three vital rates (no-niche-RGS). If the invasion growth rate of a species is negative, the species will have zero equilibrium cover. See Fig. 1 for the abbreviations for plant communities, and *Materials and methods: Summary of study sites and species selection* for the abbreviations of species names.

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that we studied, but is satisfied by a large margin. Most of the species pairs are close to equivalence on the AFD axis, and they are also surprisingly close to zero niche overlap on the ND axis. As a result, the community dynamics are very strongly stabilized and very far from neutral.

If we could extend our analysis beyond common, co-occurring species, we would certainly find a wider range of ND and AFD values. An unknown number of species in the regional pool must be excluded from our permanent quadrats because of large fitness disadvantages. Similarly, species that appear briefly in our data sets but do not persist must have fairly large fitness disadvantages along with weak niche differences, at least at the local scale. Species that persist at low densities in our quadrats might also display different patterns. If their low abundance is caused by competitive suppression, they should have relatively small niche differences with respect to the dominant species, but this weak stabilization might leave them vulnerable to stochastic extinction. On the other hand, if they occupy rare environmental niches or are limited by virulent species-specific natural enemies, they might be characterized by a combination of large niche differences and large average fitness differences, a combination more likely to ensure long-term persistence (Yenni et al. 2012).

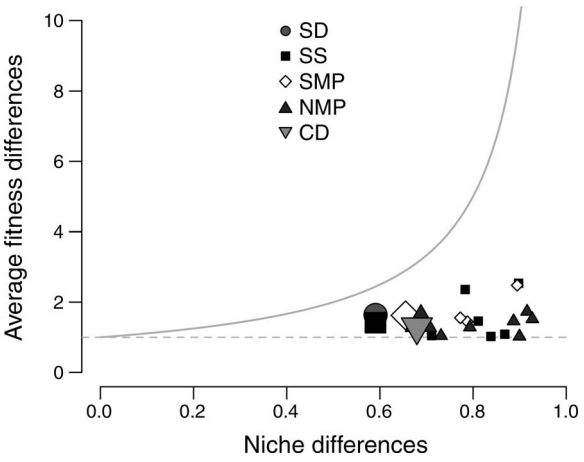


FIG. 6. Niche differences (ND) and average fitness differences (AFD) for five communities. The full community ND and AFD for each study site are represented by the larger symbols. For the three sites for which we modeled more than two species, we also show ND and AFD for pairwise species subsets of the community in smaller symbols. The gray curve was drawn using AFD = 1/(1 − ND), which marks the boundary between competitive exclusion (above and to the left of the curve) and coexistence (below and to the right). Though this coexistence criterion is accurate for the Lotka-Volterra model and MacArthur's consumer–resource model, we caution that it is an approximation for our more complex model. AFD of 1 (dashed line) means no fitness differences between species and ND of 1 means zero niche overlap. See Fig. 1 for abbreviations.

The role of recruitment processes in stabilizing coexistence

Though many theoretical studies have investigated coexistence for species with complex life histories (Moll and Brown 2008, Fujiwara et al. 2011), our work represents a unique empirical test of the relative impact of different vital rates on community dynamics. Our results indicate that while stabilizing mechanisms operate within each life stage, the recruitment process makes much larger contributions than survival or growth to fluctuation-independent coexistence. This pattern appeared in all five communities that we studied (Fig. 5). In every community, removing niche differences from recruitment processes (by perturbing the matrix of interaction coefficients) caused at least one species to go extinct. Removing niche differences from survival or growth caused far fewer extinctions. Furthermore, even when removing niche differences from recruitment did not cause extinction, it typically caused much larger reductions in invasion growth rates than did removal of niche differences from survival or growth.

We have considered three explanations for the overwhelming importance of recruitment in stabilizing coexistence. The first two explanations involve methodological artifacts. First, measurement error may inflate our estimates of the strength of intraspecific density dependence (Freckleton et al. 2006), and this bias may be most severe in our recruitment models, which depend on quadrat-level cover estimates. By contrast, our survival and growth models focus on individual genets, and should be less sensitive to biases caused by measurement error. Although the pantograph (mapping) method is quite precise relative to other methods of estimating cover, recruitment is undoubtedly influenced by plants occurring outside the mapped quadrats. To evaluate the potential for measurement error to bias our results, we quantified measurement error for the species in the sagebrush community and fit new recruitment models which accounted for this source of uncertainty. We found that measurement error had little influence on our results (Appendix D).

The second methodological explanation involves the assumptions we made to extract demographic rates from the quadrat maps. We assumed that a new recruit is a plant appearing more than 5 cm from any conspecific genet present the previous year. In other words, it is impossible for recruitment to occur within 5 cm of an established plant unless the original mappers marked a plant as a seedling (which was relatively rare). As a result, as cover increases, the potential for recruitment decreases, creating a potentially strong signal of negative density dependence in recruitment. The magnitude of this effect depends on the width of the buffer applied around established plants: the larger the buffer, the higher our estimates of survival, the lower our estimates of recruitment, and the greater the potential negative effect of cover on recruitment. To test the influence of the buffer width on our results, we explored an extreme

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case in which we repeated the entire analysis, from vital rate extraction through simulation, with the buffer width set to zero. We consider this case extreme because it undoubtedly underestimates survival and overestimates recruitment. However, the results showed that four of five communities displayed similar patterns as for the 5-cm buffer case (Appendix A). The exception came from the SD community in Arizona, where *B. eriopoda* could not persist in the zero buffer case, supporting our contention that a nonzero buffer around each polygon is more realistic. The key point is that while our choice of buffer width does affect estimated survival and recruitment rates, it did not influence our main conclusions about the predominant role of recruitment in stabilizing coexistence, or about the magnitude of niche differences and average fitness differences (Appendix A).

Our third explanation is that the strong stabilization we observed at the recruitment stage is a real biological phenomenon, not a methodological artifact, which reflects the many potential ways for niche differentiation to occur throughout the recruitment process. In contrast to survival and growth, our treatment of recruitment involves many demographic transitions: seed production, dispersal, germination, emergence, seedling survival, and establishment. Each process could be influenced by different biotic and/or abiotic factors, and the effects of these factors could be species specific. In other words, recruitment may offer more opportunities for stabilizing niche differences to operate as compared with survival and growth. In addition, many plant pathogens have stronger effects on seedlings than on mature individuals (Freckleton and Lewis 2006), and the potential for pathogens to exert species-specific effects (Janzen 1970, Connell 1971, Hersh et al. 2012) could make them a powerful source of niche differences. Finally, our results are consistent with studies from a wide range of ecosystems showing that strong intraspecific competition and plant–soil feedbacks operating at the recruitment stage play a critical role in shaping community structure and promoting coexistence (Grubb 1977, Warner and Chesson 1985, Tilman 1997, McCauley 2006, Comita et al. 2010, Mangan et al. 2010, Mordecai 2011, Martorell and Freckleton 2014).

Population ecologists may be surprised by the important role of recruitment in promoting coexistence of long-lived perennial species, given that elasticity analyses of structured population models for such species typically show that survival or growth contribute more to finite population growth rates than does recruitment (Pfister 1998, Franco and Silvertown 2004). But the apparent paradox dissolves once the differences in these analyses are considered. Our analysis focuses on invasion growth rates using density-dependent multispecies models, while the life-history studies focus on long-term population growth rates in single-species, density-independent models. In fact, Dalgleish et al. (2011) built single-species, density-independent models for our sagebrush steppe species and found that recruitment contributed less to population growth than survival or growth. Survival may contribute most to the overall population growth of long-lived species at the same time that recruitment processes contribute most to the stabilization of coexistence.

Our results are consistent with recent research in other systems

Plant coexistence studies often begin by restating the paradox of diversity (Hutchinson 1961): Since all plants compete for a few limiting resources (water, soil nutrients, and light) how is it possible for so many species to coexist in small areas? The implicit assumption is that niche overlap between species with such similar resource requirements should be high and, as a result, competitive exclusion should be common. From this perspective, our results seem hard to believe: how is it possible for co-occurring perennial grasses to have virtually zero niche overlap?

The lack of niche overlap that we observed may seem surprising, but it is actually consistent with recent findings from other natural study systems. To our knowledge, only two other empirical studies have quantified niche and fitness differences. Levine and HilleRisLambers (2009) seeded species pairs in small plots in a serpentine annual community. For the six species that coexisted in their experimental plots, niche differences spanned a similar range to the one we observed, with the minimum and maximum values just slightly lower than ours (J. HilleRisLambers and J. M. Levine, *unpublished data*). The range of fitness differences they estimated also closely matched ours, with a maximum value just slightly higher.

Narwani et al. (2013) studied a freshwater green algae community in laboratory mesocosms using Carroll et al.'s (2011) approach. They showed that for 24 of 25 pairwise comparisons, the values for niche differences were less than 0.35, and the values for average fitness differences were less than 1.4. It should not be surprising that the field studies, and especially our long-term observational study, provide evidence of stronger niche differentiation than the short-term mesocosm study. The field data sets should encompass more spatial and temporal environmental heterogeneity, more variation in the degree and type of resource limitation, and far more opportunities for interactions with predators, parasites, and other natural enemies, some of which may involve important legacy effects (i.e., plant–soil feedbacks) that would be missing from a short-term lab experiment. In fact, our study may actually underestimate the strength of niche differences because it ignores the potentially stabilizing effect of broad-scale spatial heterogeneity (Sears and Chesson 2007).

Although few studies have quantified overall niche differences, a number of recent field studies have described differences in the strength of intra- and interspecific interactions. In general, they have shown

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that intraspecific limitation is much stronger than interspecific limitation, consistent with large niche differences. In both tropical and temperate forests, seedling survival and recruitment are very sensitive to the density of conspecifics, but insensitive to the density of heterospecifics (Comita et al. 2010, 2014, Metz et al. 2010, Alvarez-Loayza and Terborgh 2011, Johnson et al. 2012, Piao et al. 2013, Jansen et al. 2014). Farrer and colleagues (Farrer et al. 2010, Farrer and Goldberg 2011) found that conspecific interactions were more intense than heterospecific interactions in a perennial grassland. Finally, plant–soil feedback experiments have shown that conspecific feedbacks are more negative than heterospecific feedbacks (reviewed by Kulmatiski et al. 2008).

The accumulating evidence for large niche differences should refocus attention on longstanding questions about the underlying mechanisms. How, exactly, do plant species manage so successfully to avoid competing with each other? Recent answers to this question have highlighted the role of species-specific natural enemies, especially soil pathogens (e.g., Bagchi et al. 2010, 2014). The importance of recruitment processes in driving niche differences in our own analysis is also suggestive of natural enemies, since seeds and seedlings are so vulnerable to them. The natural enemies explanation may be compelling because it is easier for us to imagine many species-specific enemies than many different ways that species could partition limited resources in space and time. However, this preference may reflect our own limitations more than nature's. If we consider high-dimensional interactions among many resources and environmental factors, the potential for niche differences to emerge from these factors also may be quite high (Clark et al. 2010).

Future directions

Species-specific responses to temporal environmental variation represent one important class of coexistence mechanism (Chesson 2000). When we removed such fluctuation-dependent mechanisms by simulating our models in a constant environment, we found relatively little effect on invasion growth rates and equilibrium abundances (Fig. 5). The temptation is to conclude that fluctuation-dependent mechanisms are weak in these ecosystems (e.g., Adler et al. 2010). However, we now believe that the models we fit may not be appropriate for rigorously quantifying fluctuation-dependent mechanisms. For example, the temporal storage effect requires positive covariance between the effects of competition and the environment (Chesson and Huntly 1989): in years most favorable to a species, it should also be the most vulnerable to competition. However, we did not specify our vital rate regressions in a way that makes the covariation between the environment (or random year effects) and competition explicit. As a result, our model may not allow full expression of a temporal storage effect and the simulations may underestimate its strength. Such differences in model specification may explain why a previous analysis, based on a very different modeling approach, showed a strong storage effect in the Kansas community (Adler et al. 2006) in contrast to the weak effect we observed here. We plan to address this model specification issue in future work focused on the population and community consequences of environmental variability.

A second important direction for future research is incorporating facilitation into coexistence theory. Many studies have demonstrated that positive interactions (facilitation) among organisms are common and can have a strong impact on community dynamics (Callaway et al. 2002, Callaway 2007, Martorell and Freckleton 2014, McIntire and Fajardo 2014). We found that some of our study species achieved higher low-density growth rates in the presence of resident vegetation than when grown without competitors. These facilitative interactions prevented us from applying either Chesson's (2013) or Carroll et al.'s (2011) approaches for estimating niche and fitness differences. In order to apply the current theory, we refit our models, constraining interactions to be negative. For the communities we studied, the facilitative interactions were weak enough that excluding them had little impact on simulated community dynamics (Appendix C), but for communities in which facilitation is strong, simply ignoring it will not be acceptable (Venail et al. 2014). Clearly, we need new theory to allow us to estimate niche and fitness differences in the presence of positive interactions. Doing so may require moving from phenomenological representations of species interactions to mechanistic models (Holland and DeAngelis 2010), which will pose a challenge if we hope to avoid a loss of generality.

The strong stabilization we observed occurs because the population growth of each species is much more sensitive to variation in the local abundance of conspecifics than heterospecifics. As a result, our models predict that removal of a common species would not lead to a dramatic competitive release in competing common species, which contrasts sharply with the zero-sum dynamics assumed (but not required) by many neutral models (e.g., Hubbell 2001). The implications extend to applied ecology as well. For example, the relative importance of intra- and interspecific interactions should determine the magnitude of indirect effects of climate change mediated by species interactions, a topic of great current interest (Brooker 2006, Gilman et al. 2010, Adler et al. 2012, 2013, Angert et al. 2013, Buckley 2013). Indirect effects should be small when niche differences are large (Adler et al. 2012). Based on recent results from other systems, the dynamics we have observed do not appear to be confined to water-limited plant communities. They do, however, challenge the traditional focus of community ecology on strong resource competition. The emerging evidence suggests that intense interspecific competition may not be the

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primary driver of dynamics among common, co-occurring species, and should inspire renewed efforts to identify species-specific differences in the sources of intraspecific density dependence.

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Supplemental Material

Ecological Archives

Appendices A–I are available online: http://dx.doi.org/10.1890/14-1741.1.sm

Data Availability

Data and computer code used to fit the vital rates models and to run simulations have been deposited in Dryad: http://dx.doi.org/10.5061/dryad.37n6f.

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